

Bridging the Scale Gap: Cross-View Localization from Dense Rover LiDAR to Coarse Lunar DEMs

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Abstract—Rovers on the Moon must recover their absolute pose without GNSS, and orbital Digital Elevation Models (DEMs) are, in practice, the primary globally available geometric prior before landing. Matching onboard LiDAR against an orbital DEM offers an absolute, illumination-invariant position fix, yet to our knowledge, existing methods have not been evaluated in this lunar cross-view setting using real orbital topography. We benchmark four representative geometric and learning-based methods—DARCES, BEVPlace++, RING, and ICP—on a simulated lunar dataset built from real orbital DEMs of five sites, using single-frame LiDAR queries without odometry across a DEM-resolution sweep from 0.25 to 10 m/pixel. Meaningful localization is achieved only on sites with distinctive and spatially unique terrain structure, and with relatively fine DEMs; performance remains poor on feature-poor mare terrain regardless of resolution and degrades sharply at the coarser resolutions that cover most of the lunar surface. These findings identify two key bottlenecks for globally applicable lunar cross-view localization: robustness to coarse DEM resolution and robustness to feature-poor terrain.

I. INTRODUCTION

Sustained robotic activity on the Moon requires a rover to recover its absolute pose on a globally consistent map. GNSS is unavailable on the Moon, wheel odometry drifts over long traverses, and vision-based methods struggle with extreme grazing-angle illumination and low-texture regolith. Orbital Digital Elevation Models (DEMs) derived from LRO instruments such as LOLA and LROC [1], [2] are a principal globally georeferenced geometric prior available before landing. Registering onboard LiDAR geometry against such a DEM yields an absolute, illumination-invariant position fix (Fig. 1), making cross-view LiDAR-to-DEM localization a natural building block for lunar autonomy.

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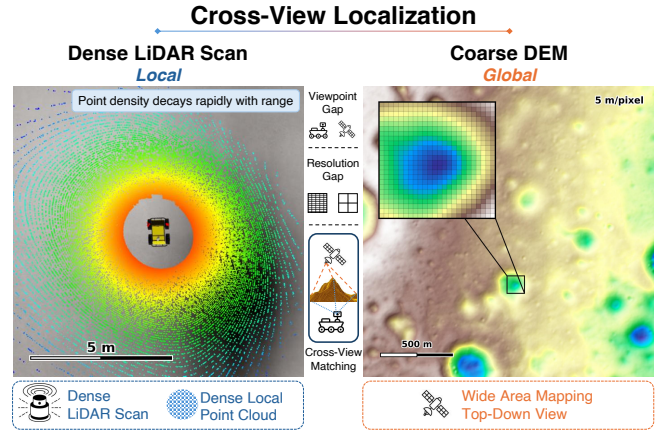


Fig. 1: **Cross-view LiDAR-to-DEM localization.** A dense rover LiDAR scan is registered to a coarse orbital DEM to recover the rover’s global pose. Both panels are rendered from our OmniLRS simulation environment used in the experiments.

Candidate methods for this task differ in design intent and validation environment. A classical planetary method [3] correlates rover LiDAR scans with orbital elevation maps and was validated on terrestrial lunar-analog terrain. Terrestrial methods, including learning-based LiDAR place recognition [4] and handcrafted geometric methods that jointly estimate place, orientation, and translation [5], were developed on urban driving data. None has been evaluated on matching a rover LiDAR scan to a coarse orbital DEM built from real lunar topography.

This paper provides that evaluation. We benchmark these methods, together with ICP [6] as a naive direct-registration baseline, on a shared set of lunar query scans. We retain each method’s original localization pipeline and apply only the preprocessing required for evaluation. We evaluate at five sites of diverse terrain morphology across a controlled DEM-resolution sweep, distinguishing source orbital DEMs from simulation-derived reference DEMs.

II. EVALUATED METHODS

A. DARCES

DARCES [3] is the feature-matching core of the MOGA pipeline proposed by Carle et al. for long-range rover localization against orbital elevation maps. Both the rover LiDAR scan and the orbital map are converted into 2.5-D elevation grids, from which topographic features are detected by

TABLE I: **Source DEMs and localization-relevant characteristics of the selected lunar sites.** Traverse length denotes the total rover trajectory used for data acquisition at each site.

Site	Source DEM	Source res. ($m/pixel$)	Traverse length (m)	Morphology and benchmark role
Apollo 11	NAC DTM	2.0	433.6	Low-relief mare terrain; feature-poor baseline.
Apollo 14	NAC DTM	2.0	2723.6	Bowl-shaped crater spanning floor, wall, and rim; compact crater landmark.
Apollo 15	NAC DTM	2.0	3071.1	Plateau adjacent to Hadley Rille; elongated directional terrain cue.
Apollo 17	NAC DTM	1.5	738.0	Relatively flat valley floor bounded by steep massifs; contrast between local and map-scale relief.
Haworth	NAC DTM	1.0	316.7	Broad depressions and irregular slopes; distributed relief without a dominant landmark.

morphological dilation. DARCES generates pose hypotheses from triplets of local and global features with geometrically compatible pairwise distances, verifies them by terrain agreement, and outputs a 2.5-D pose (planar position, heading, and vertical offset). Consistent with our single-frame protocol, we omitted the subsequent RANSAC and multi-frame odometry-compensated refinement and evaluated only the per-scan DARCES stage. DARCES may return no estimate when insufficient compatible features are found.

B. BEVPlace++

BEVPlace++ [4] is a learning-based LiDAR global localization method that projects a LiDAR scan into a bird’s-eye-view (BEV) density image and extracts rotation-invariant features using a rotation-equivariant network. It performs place recognition by retrieving the most similar reference BEV image and subsequently estimates the 3-DoF pose of the query relative to the retrieved map. In our setting, reference BEV images were rendered from the DEM, and query BEV images were generated from rover LiDAR scans. Because the publicly released weights were trained on terrestrial urban data, we fine-tuned the model on lunar LiDAR–DEM query–reference pairs from our simulated dataset and used this fine-tuned model for evaluation. The data used for fine-tuning were excluded from evaluation.

C. RING

RING [5] is a geometric LiDAR global localization method that represents a point cloud as a Radon sinogram. It constructs a translation-invariant descriptor using the Fourier magnitude of the sinogram and performs place recognition and relative yaw estimation through circular cross-correlation. The relative translation is subsequently recovered from the angle-dependent shifts in the sinogram. In our setting, we generated query representations from rover LiDAR scans and reference representations from DEM-derived point clouds. For preprocessing, we retained points satisfying $z > -0.6$ m relative to the LiDAR sensor origin for both query and reference point clouds, using an empirically selected threshold held fixed across all datasets and DEM resolutions.

D. ICP Baseline

Iterative Closest Point (ICP) [6] is a classical point-cloud registration method based on nearest-neighbor correspondences. Since ICP is sensitive to initialization and may converge to local minima, we adopted a simple multi-start strategy [7]. For each query scan, we uniformly sampled

10 initial x - y positions within the predefined evaluation region and set the initial roll, pitch, and yaw to zero. We independently performed ICP from each initialization against the DEM point cloud at each evaluated resolution and selected the hypothesis with the highest registration fitness as the final localization estimate. We reused the same initial x - y hypotheses across all DEM resolutions to ensure a fair comparison across DEM resolutions.

III. EVALUATION

A. Lunar Simulation Dataset

We constructed a simulated lunar localization dataset using OmniLRS [8] to obtain LiDAR observations with synchronized ground-truth poses and procedural crater augmentation. The dataset consists of rover LiDAR scans and reference DEMs at multiple grid spacings. Each environment was instantiated from a real orbital DEM: the Apollo 11, 14, and 15 terrains use LROC NAC DTMs [9], Apollo 17 uses an LROC NAC DTM of the Taurus–Littrow Valley [10], and Haworth uses a shape-from-shading DEM [11].

The five sites were selected to span localization-relevant morphology classes rather than as a statistically representative sample of the lunar surface. As summarized in Table I, the sites vary in spatial scale, directional structure, and distinctiveness of their topographic cues, enabling us to examine how terrain morphology interacts with reference-DEM grid spacing.

Fig. 2 shows the terrain morphology of the selected regions together with the rover trajectories used for data acquisition, illustrating the diversity covered by the dataset.

Query LiDAR data. Predefined trajectories were designed to sample the characteristic terrain morphology of each site, and the rover was driven along these trajectories in the physics-based simulation during data acquisition. To decouple localization data acquisition from rover mobility constraints, data for Apollo 14 were collected along multiple feasible trajectory segments spanning the crater floor, wall, and rim. A Husky-based rover carried a 128-channel Ouster OS1 REV7 RTX LiDAR operating at 20 Hz with 1,024 azimuth columns, 360° horizontal field of view (FoV), and up to two returns per ray. Scans were synchronized with simulator ground truth using the LiDAR timestamp and subsampled at approximately 1 m trajectory intervals.

Reference DEMs. Separately, each terrain was reconstructed in OmniLRS and augmented with procedural craters. The resulting terrain was exported at 0.25 $m/pixel$ to

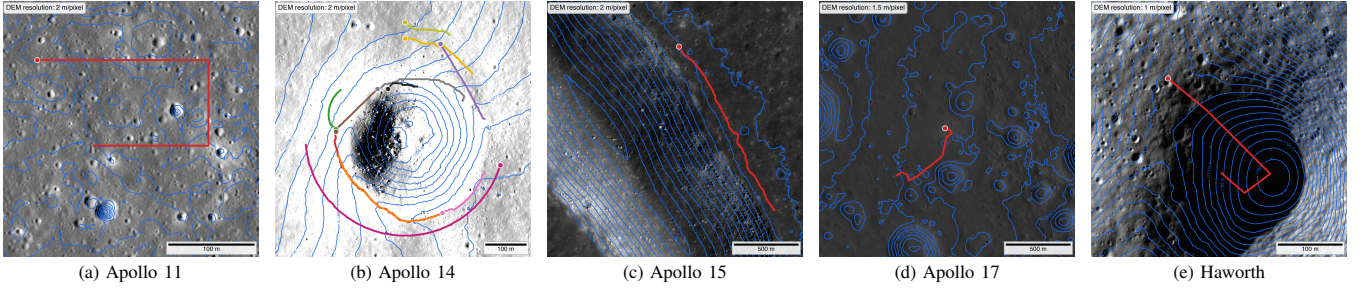


Fig. 2: **Lunar dataset sites and rover trajectories.** Selected lunar terrain regions used for LiDAR data acquisition. Red curves denote rover trajectories, blue curves denote elevation contours, and the different colored curves in Apollo 14 indicate separate feasible trajectory segments. Source DEM resolution and spatial scale are annotated in each panel.

preserve these added features in the reference map, after which spatially aligned DEMs at 0.5, 1, 2, 5, and 10 $m/pixel$ were generated by downsampling.

Evaluation metrics. For each query, the horizontal localization error is $e_{xy} = \sqrt{(\hat{x} - x)^2 + (\hat{y} - y)^2}$. We report the median, RMSE, and localization success rate within 5m (SR@5m). The 5m threshold, corresponding to one grid interval of a 5m/pixel DEM, is fixed across all resolutions as an absolute benchmark tolerance. Missing estimates are counted as failures.

B. Localization Performance across DEM Resolutions

Table II and Fig. 3 summarize localization performance across all sites and DEM resolutions. BEVPlace++ and RING generally degraded as the DEM became coarser, most clearly on Apollo 14 and Apollo 15, while Haworth showed a weaker but consistent decline for RING. On Apollo 14, both methods maintained 74–85% SR@5m between 0.25 and 2 $m/pixel$ but degraded sharply at 5–10 $m/pixel$, indicating that coarse DEMs remove discriminative terrain geometry. Similar behavior on Apollo 15 suggests that grid spacings of roughly 1–2 $m/pixel$ preserve much of the geometry required for localization at these structured sites.

The site-wise results indicate that terrain morphology, rather than resolution alone, governs localization performance. Apollo 14 and Apollo 15 were the most favorable sites, with their crater and rille structures providing spatially distinctive relief observable from single rover scans and supporting up to 91% SR@5m at fine resolutions. In contrast, the low-relief Apollo 11 remained unreliable at all resolutions. Apollo 17 provides a counterexample to the assumption that large global relief alone is sufficient: prominent map-scale relief does not necessarily contribute to localization when it is not consistently observable from the rover viewpoint. Haworth exhibited intermediate performance due to its distributed relief without a dominant landmark. Overall, useful localization cues must be both spatially distinctive and observable within the local rover LiDAR footprint.

Method-specific behaviors are also notable. DARCES frequently failed to return a pose estimate at the finest or the coarsest resolution, while its estimate count and SR@5m peaked at intermediate grid spacings. RING was competitive with or superior to BEVPlace++ on Apollo 14, 15,

and Haworth, yet performed poorly across all resolutions on Apollo 17, with its SR@5m reaching only 3.6% at 5 $m/pixel$. This behavior may reflect a mismatch between the valley geometry and RING’s fixed height-filtering threshold. ICP failed almost uniformly regardless of resolution. These results suggest that ICP is insufficient to bridge the cross-view gap in this setting.

Finally, several fine-resolution conditions produced small median errors but considerably larger RMSE. The large median–RMSE gap indicates substantial outliers: many queries are localized accurately, while a subset incurs very large errors. Median position error alone can therefore overstate practical reliability, and a deployable pipeline would require outlier rejection or multi-hypothesis verification.

IV. CONCLUSION

Our evaluation shows that existing localization methods can produce meaningful results when the terrain contains distinctive topographic structure and a relatively fine DEM is available as a prior map. However, such conditions occur only in limited regions of the Moon: while some regions are covered by NAC DTMs at 1–2 $m/pixel$ [9], most of the lunar surface is covered only by DEMs at 5 $m/pixel$ or coarser. Robustness across DEM resolutions is therefore essential for localization across the lunar surface, rather than only at well-mapped sites. Moreover, all evaluated methods performed poorly on feature-poor terrain lacking distinctive structures such as large craters or rilles, indicating that reliance on salient topographic features is itself a limiting factor.

In future work, we aim to develop a cross-view localization method that remains robust across a wide range of DEM resolutions and remains reliable on terrain without distinctive topographic features. In addition, RING and BEVPlace++ require generating pseudo-LiDAR point clouds over the entire DEM to construct reference representations, which incurs substantial computational cost; a method that operates directly on the DEM, or requires far less preprocessing, would be desirable.

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TABLE II: **Localization performance across DEM resolutions.** Median and RMSE denote horizontal position errors, and SR@5m is the success rate within 5 m. #Est. denotes the number of estimates returned by DARCES; missing estimates are counted as failures. The best and second-best SR@5m per row are highlighted in **dark** and **light** green.

Dataset	DEM res. (m/pixel)	DARCES [3]				BEVPlace++ [4]			RING [5]			ICP [6]		
		Median (m)	RMSE (m)	SR@5m (%)	#Est.	Median (m)	RMSE (m)	SR@5m (%)	Median (m)	RMSE (m)	SR@5m (%)	Median (m)	RMSE (m)	SR@5m (%)
Apollo 11	0.25	—	—	0.0	0/100	101.25	116.59	15.0	58.06	88.90	24.0	84.97	95.74	1.0
	0.5	5.63	4.95	1.0	3/100	101.86	120.01	15.0	50.81	85.81	29.0	83.30	95.55	1.0
	1	3.53	65.90	12.0	18/100	101.99	107.54	13.0	51.68	88.25	28.0	85.49	95.74	0.0
	2	5.49	88.27	12.0	26/100	102.14	123.10	10.0	59.71	91.69	26.0	85.59	97.08	0.0
	5	5.84	48.08	15.0	30/100	104.55	125.31	10.0	66.86	94.18	23.0	82.95	92.98	1.0
	10	40.54	74.75	0.0	9/100	102.98	124.78	1.0	65.75	97.35	5.0	95.88	107.23	1.0
Apollo 14	0.25	2.64	2.67	2.0	2/100	1.49	110.09	78.0	1.66	143.25	74.0	52.02	108.29	0.0
	0.5	2.35	2.89	37.0	38/100	1.40	90.94	84.0	1.48	144.07	77.0	52.03	108.29	0.0
	1	2.86	101.40	40.0	44/100	1.61	70.45	85.0	1.68	133.53	80.0	52.27	107.62	0.0
	2	3.54	59.77	22.0	31/100	2.33	87.49	85.0	2.42	129.19	79.0	52.24	109.53	1.0
	5	5.36	6.48	9.0	24/100	41.44	182.63	28.0	8.40	181.05	25.0	63.79	109.69	0.0
	10	—	—	0.0	0/100	220.79	233.24	0.0	171.82	228.30	1.0	91.44	126.13	2.0
Apollo 15	0.25	1.39	424.33	35.0	43/100	0.89	221.20	82.0	1.23	249.38	88.0	570.67	642.95	0.0
	0.5	1.52	265.89	71.0	84/100	0.90	256.00	83.0	1.23	228.29	90.0	557.28	640.73	0.0
	1	1.54	116.63	88.0	97/100	1.12	234.34	84.0	1.46	233.47	87.0	574.82	640.90	0.0
	2	1.99	268.13	80.0	94/100	2.06	338.72	71.0	1.78	158.93	91.0	612.56	662.97	0.0
	5	4.69	404.96	46.0	83/100	418.10	620.57	19.0	4.55	399.26	55.0	557.44	642.33	0.0
	10	318.65	487.21	0.0	58/100	548.23	665.54	0.0	458.35	561.76	2.0	535.28	650.40	0.0
Apollo 17	0.25	—	—	0.0	0/28	12.39	222.58	39.3	219.76	269.87	0.0	272.39	317.33	0.0
	0.5	3.45	3.45	3.6	1/28	21.94	229.21	42.9	247.61	283.15	0.0	272.41	317.33	0.0
	1	364.85	547.21	7.1	7/28	46.12	191.98	32.1	224.98	256.46	0.0	272.46	317.29	0.0
	2	68.41	379.16	25.0	18/28	67.02	233.70	39.3	241.34	274.93	0.0	272.74	317.29	0.0
	5	4.96	272.95	39.3	21/28	82.62	219.13	28.6	241.86	254.26	3.6	300.77	327.72	0.0
	10	506.08	650.36	10.7	15/28	214.15	233.00	0.0	228.75	268.42	0.0	288.62	314.83	0.0
Haworth	0.25	—	—	0.0	0/100	44.64	65.52	8.0	16.69	71.76	31.0	92.62	102.00	0.0
	0.5	12.88	95.31	4.0	19/100	59.24	71.54	7.0	22.18	76.87	28.0	92.64	101.63	0.0
	1	52.38	172.69	4.0	44/100	53.99	67.78	5.0	20.47	76.76	27.0	92.66	101.65	0.0
	2	145.35	157.66	1.0	41/100	58.84	72.88	2.0	28.10	77.89	22.0	90.43	101.68	0.0
	5	227.54	220.88	0.0	6/100	60.79	81.72	2.0	48.60	87.23	14.0	92.43	102.50	0.0
	10	—	—	0.0	0/100	63.98	84.39	3.0	50.82	95.07	3.0	87.41	98.58	0.0

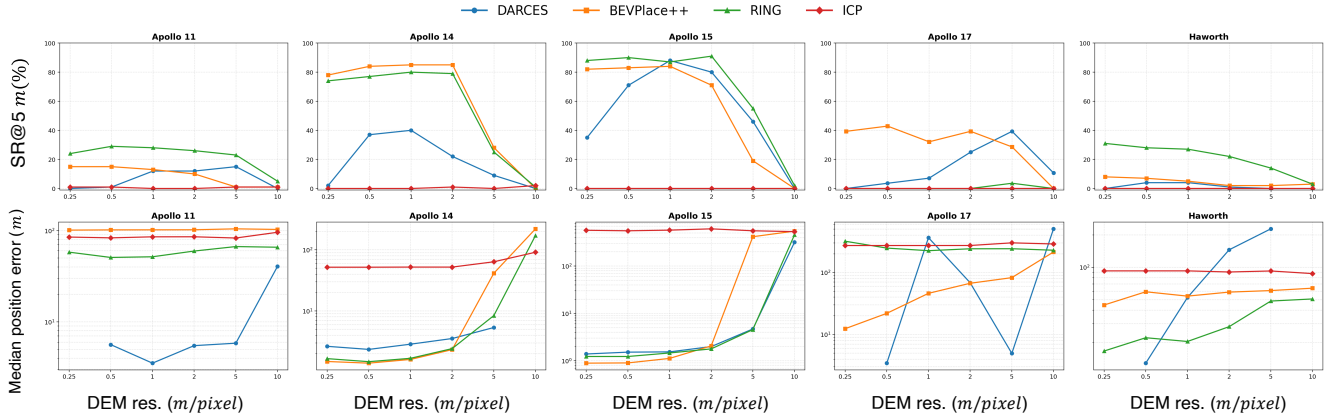


Fig. 3: **Localization performance across DEM resolutions.** Top: SR@5m, Bottom: median position error on a log-scale.

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